Evidence gap map of bias–mitigation intersections in histopathology AI fairness research

Mitigation family (grouped)

Bias category	Pre-processing		Feature-level		Protocol / Data		Collaborative		Domain Adaptation		Foundation / Resampling		Post-processing		Total studies (per bias)
	Stain normalisation	Data augmentation	ComBat / Feature harmonisation	Fairness-aware training	Preserved-site CV	Generative augmentation	Federated learning	Causal methods	Domain adaptation	Adversarial debiasing	SSL / Foundation models	Resampling	Threshold optimisation	Explanation constraints	
1 Technical / Domain shift	10 ●	8 ▲	2 ★	1 ★	1 ★	2 ▲	0 –	1 ▲	7 ●	4 ▲	4 ▲	0 –	0 –	0 –	40
2 Selection / Sampling bias	1 ●	1 ▲	1 ★	1 ★	3 ★	0 –	2 ★	1 ▲	0 –	0 –	1 ▲	0 –	0 –	0 –	11
3 Institutional / Batch bias	4 ●	1 ●	2 ★	1 ★	2 ★	0 –	3 ★	1 ▲	1 ●	2 ▲	1 ▲	0 –	0 –	0 –	18
4 Class / Label bias	2 ●	1 ▲	0 –	0 –	0 –	1 ▲	0 –	0 –	0 –	0 –	1 ▲	1 ●	0 –	2 ▲	8
5 Confounding bias	1 ●	0 –	1 ★	0 –	3 ★	0 –	1 ★	2 ▲	0 –	1 ▲	1 ▲	0 –	0 –	0 –	10
6 Representation bias	0 –	0 –	0 –	0 –	0 –	2 ▲	1 ★	0 –	0 –	0 –	1 ▲	0 –	0 –	0 –	4
7 Demographic bias	0 –	0 –	0 –	3 ★	1 ★	1 ▲	0 –	0 –	0 –	0 –	0 –	1 ●	2 ▲	0 –	8
8 Algorithmic bias	0 –	0 –	0 –	0 –	0 –	0 –	0 –	1 ▲	0 –	0 –	0 –	0 –	1 ●	0 –	2
Total studies (per mitigation)	**18**	**11**	**6**	**6**	**10**	**6**	**7**	**6**	**8**	**7**	**9**	**2**	**3**	**2**	

Number of studies (log-scaled)

0 | 1–2 | 3–5 | 6–10 | ≥11

Highest effectiveness grade achieved

- ★ **Strong:** ≥3 positive studies; consistent fairness gains
- ▲ **Moderate:** 2+ positive studies; minor trade-offs
- ● **Mixed/Failed:** contradictory or no improvement
- – **None:** no studies tested this pair

Note

Each cell shows the number of unique studies that explicitly tested the bias–mitigation pair and the highest effectiveness grade across those studies (symbol). Data from systematic review of 78 studies (Jan 2017 – Mar 2026). Totals indicate studies per bias category and per mitigation family.